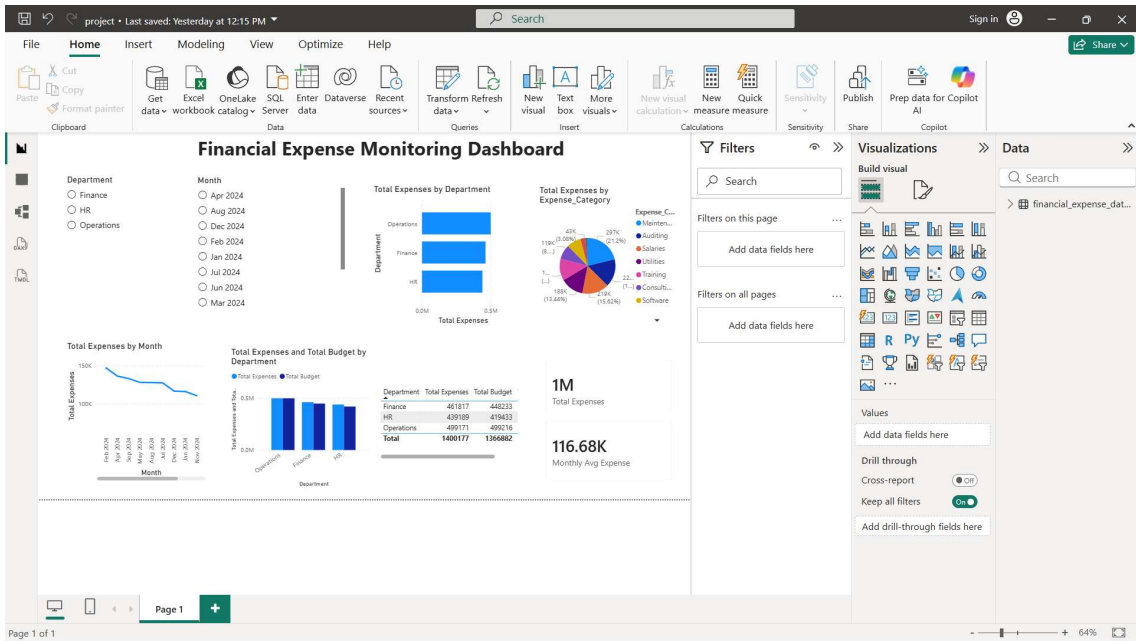


FINANCIAL EXPENCE MONITORING DASHBOARD



Question

1. Identify missing or inconsistent expense data.

During data preparation, the dataset was reviewed for missing values, incorrect entries, and inconsistencies. Columns such as Date, Department, Expense Category, Amount, and Budget were checked to ensure completeness. Data types were standardized (for example, converting Date into proper date format), and duplicates or blank entries were verified. No major missing values were found in key fields, and numeric fields like Amount and Budget were validated to ensure logical consistency. This step ensured reliable analysis and accurate visual outputs in the dashboard.

BEFORE:

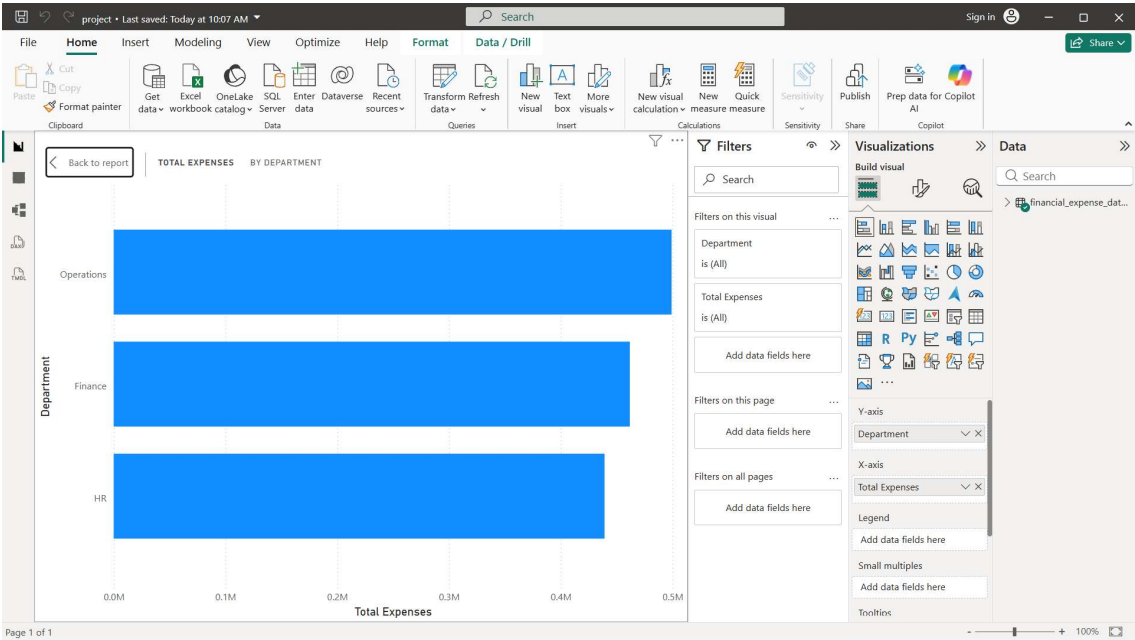
	Date	Department	Expense_Category	Amount	Budget	Vendor
1	06-04-2024	Finance		41904		null
2	20-01-2024	Finance		null	-500	PowerGrid
3	26-03-2024	Finance	Salaries	23261	-500	Oracle
4	20-02-2024			null		Oracle
5	20-05-2024		Salaries	20558	null	Oracle
6	18-04-2024	Finance		42315	-500	SAP
7	14-01-2024	Finance		42315	42158	
8	04-03-2024	Finance	Software	46819	null	
9	02-02-2024	Finance		44476	26613	
10	20-04-2024	HR	Salaries	2453	-500	SAP
11	19-01-2024	Operations	Software	null	3957	PowerGrid
12	16-03-2024		Salaries	null	16096	Oracle
13	17-05-2024	Operations		20750	14278	SkillUp
14	30-05-2024	Finance		null		Oracle
15	16-03-2024		Maintenance	7742	23150	SAP
16	03-06-2024			null		SAP
17	16-04-2024	Operations	Salaries	49950	-500	SkillUp
18	30-03-2024	Finance	Training	null	37227	SAP
19	28-01-2024	Operations	Salaries	null		SAP
20	27-05-2024	Finance		26533	-500	SAP
21	28-06-2024	HR	Maintenance	38432	null	SAP
22	25-06-2024	Finance	Salaries	null	12987	PowerGrid
23	14-02-2024	Operations		31550	null	
24	19-04-2024	Operations	Training	null	2700	
25	20-01-2024	HR		null	-500	
26	25-04-2024	Finance	Salaries	21793	null	SkillUp
27	05-04-2024	Training		null	-500	Oracle

AFTER:

	Date	Department	Expense_Category	Amount	Budget	Vendor
1	01-01-2024	HR	Recruitment	4141		1155 Oracle
2	08-01-2024	Operations	Maintenance	33060	32115	SAP
3	15-01-2024	HR	Training	16400	18130	PowerGrid
4	22-01-2024	HR	Salaries	13320	9742	SAP
5	29-01-2024	HR	Training	49017	52517	SkillUp
6	05-02-2024	HR	Salaries	36285	35331	PowerGrid
7	12-02-2024	Operations	Utilities	42034	48421	SkillUp
8	19-02-2024	Finance	Auditing	47920	43195	Deloitte
9	26-02-2024	HR	Training	20861	19731	Oracle
10	04-03-2024	Operations	Utilities	40691	37763	Deloitte
11	11-03-2024	Operations	Utilities	17147	19039	Oracle
12	18-03-2024	HR	Recruitment	10400	14335	SAP
13	25-03-2024	Finance	Software	13486	17119	Oracle
14	01-04-2024	Operations	Utilities	18701	16046	SkillUp
15	08-04-2024	Operations	Maintenance	44219	45906	SkillUp
16	15-04-2024	Finance	Auditing	37392	37759	KPMG
17	22-04-2024	Finance	Auditing	13930	11431	KPMG
18	29-04-2024	Finance	Auditing	22074	19807	KPMG
19	06-05-2024	Finance	Consulting	25109	20910	Fixit Ltd
20	13-05-2024	Operations	Maintenance	43884	46367	PowerGrid
21	20-05-2024	Finance	Software	35543	39826	HirePro
22	27-05-2024	HR	Salaries	23423	19689	SkillUp
23	03-06-2024	Operations	Maintenance	20352	17489	SkillUp
24	10-06-2024	HR	Salaries	8711	9137	SkillUp
25	17-06-2024	Operations	Utilities	15560	18250	Fixit Ltd
26	24-06-2024	HR	Salaries	34097	37976	Deloitte
27	01-07-2024	HR		13216	8952	Deloitte

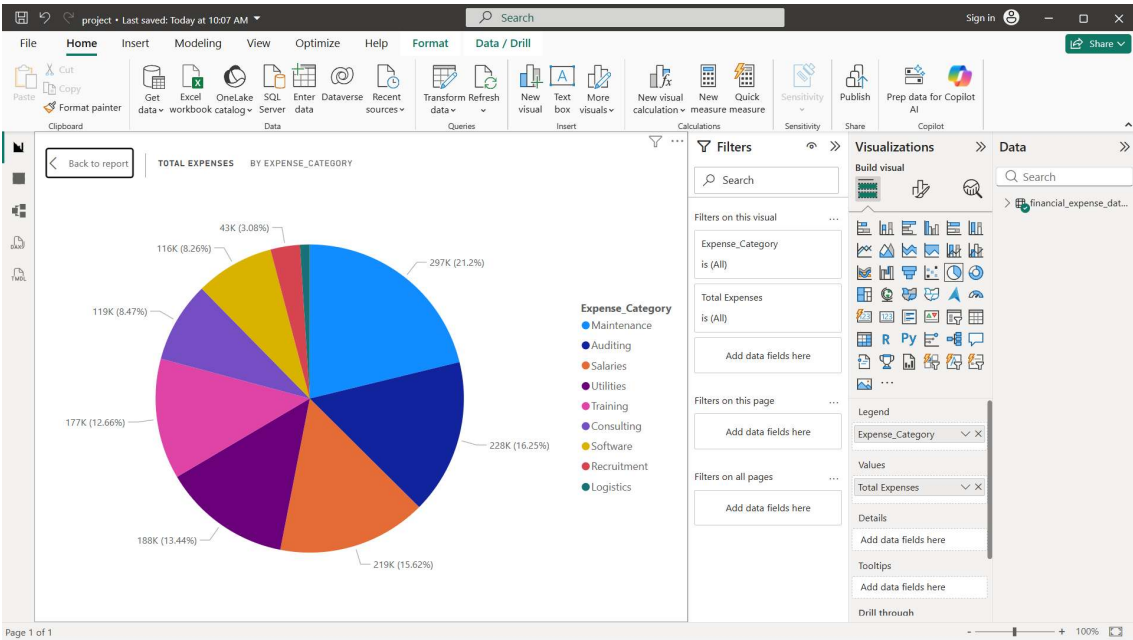
2. Which departments incur the highest total expenses?

Based on the department-wise expense analysis using a bar chart, the Operations department incurred the highest total expenses, followed by Finance and HR. This indicates that operational activities such as logistics, maintenance, and utilities contribute significantly to organizational spending. Identifying high-expense departments helps management prioritize monitoring and cost optimization.



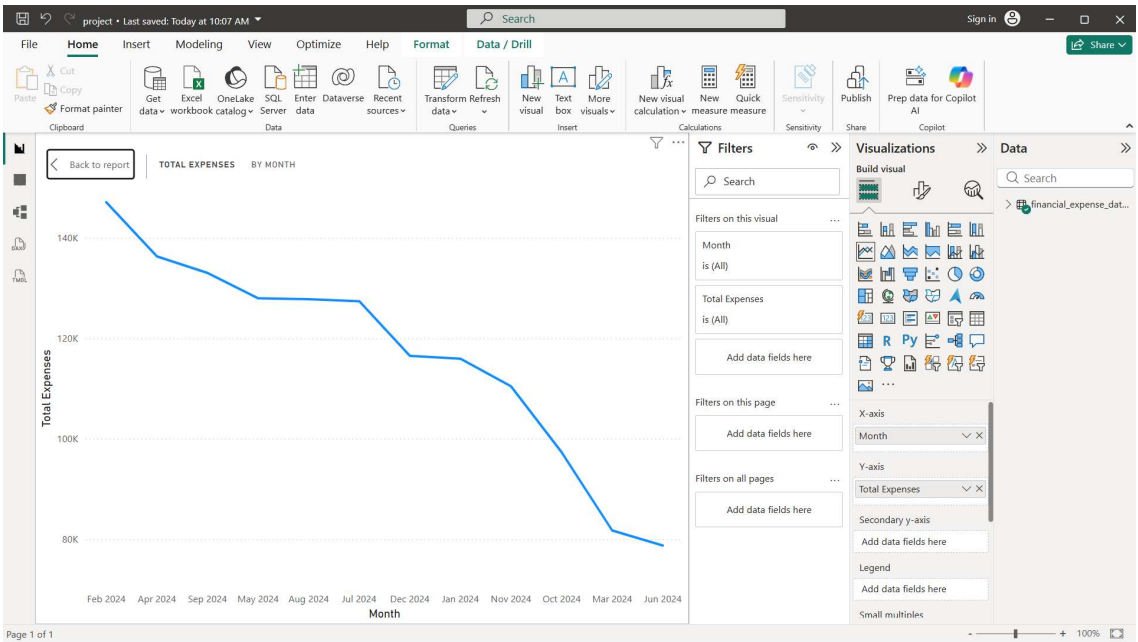
3. Analyze expense distribution across expense types

A pie chart visualization was used to examine how expenses are distributed across categories such as Salaries, Software, Maintenance, Consulting, and Utilities. The analysis showed that some categories contribute a larger share of the total expenditure compared to others. This distribution highlights key spending drivers and helps decision-makers understand where resources are being allocated. It also assists in identifying areas where optimization or reallocation may be required.



4. How do monthly expenses trend over time?

The monthly expense trend line chart revealed fluctuations in spending throughout the year. Overall, the trend shows variations across months, indicating periods of higher or lower operational costs. Monitoring these patterns enables organizations to forecast future expenses, detect anomalies, and plan budgets more effectively. Trend analysis is essential for long-term financial planning and cost control

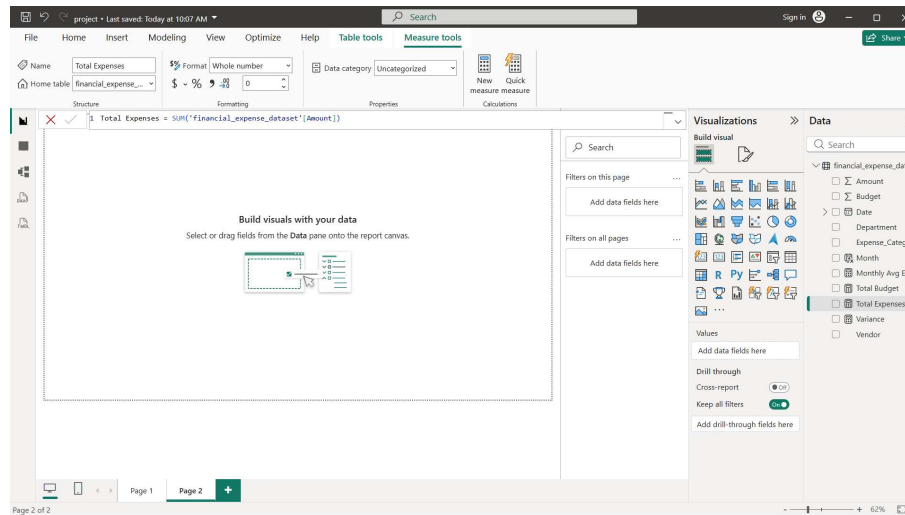


5. Create DAX measures for Total Expenses and Monthly Average Expense

Two measures were created:

1.Total Expenses was calculated using aggregation of all expense values to provide an overall spending indicator.

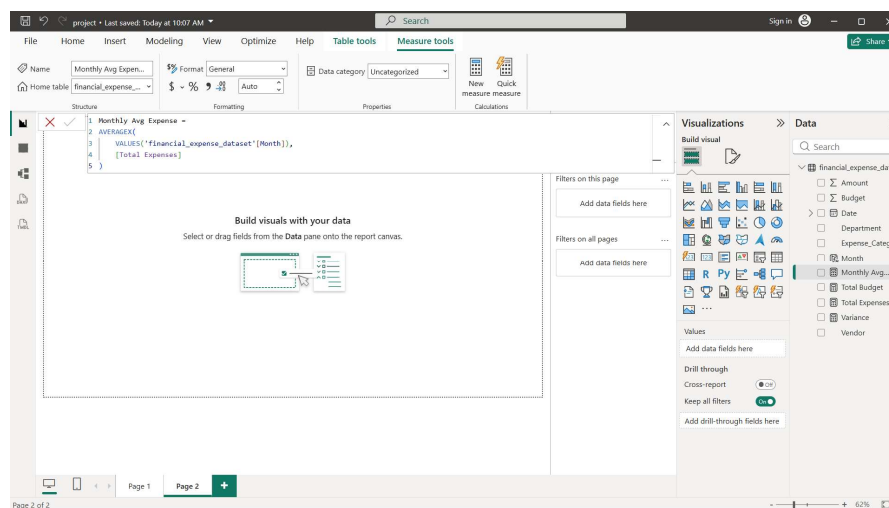
Total Expenses = SUM(financial_expense_dataset[Amount])



2.Monthly Average Expense was calculated to measure the average spending across months, giving a normalized view of expenditure trends.

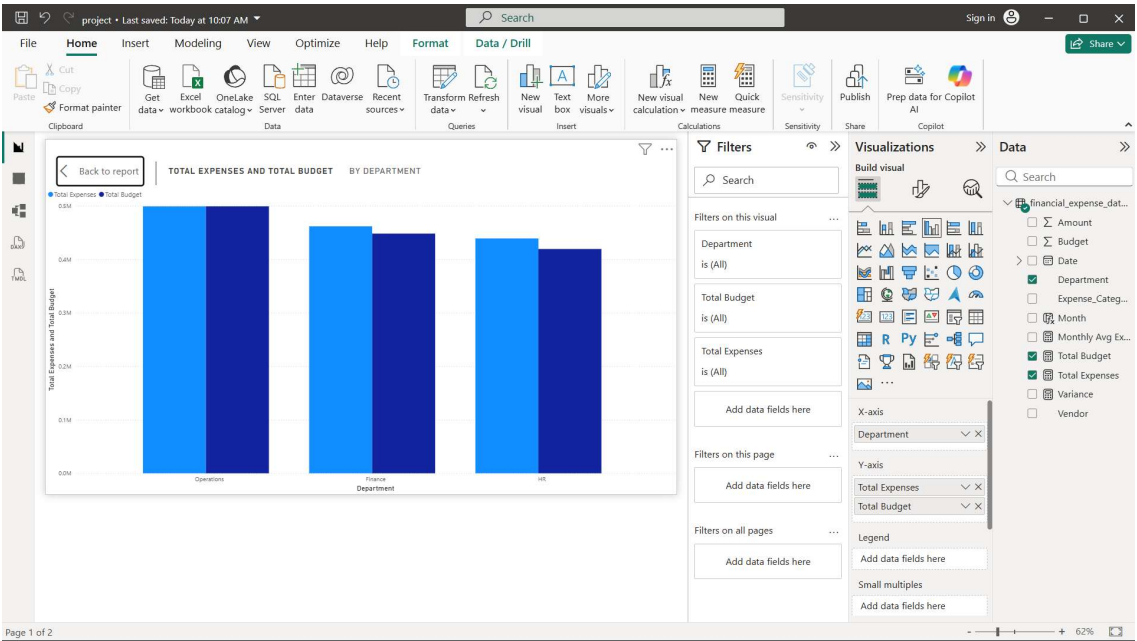
Monthly Avg Expense =

AVERAGEX(
VALUES(financial_expense_dataset[Month]),
[Total Expenses]
)



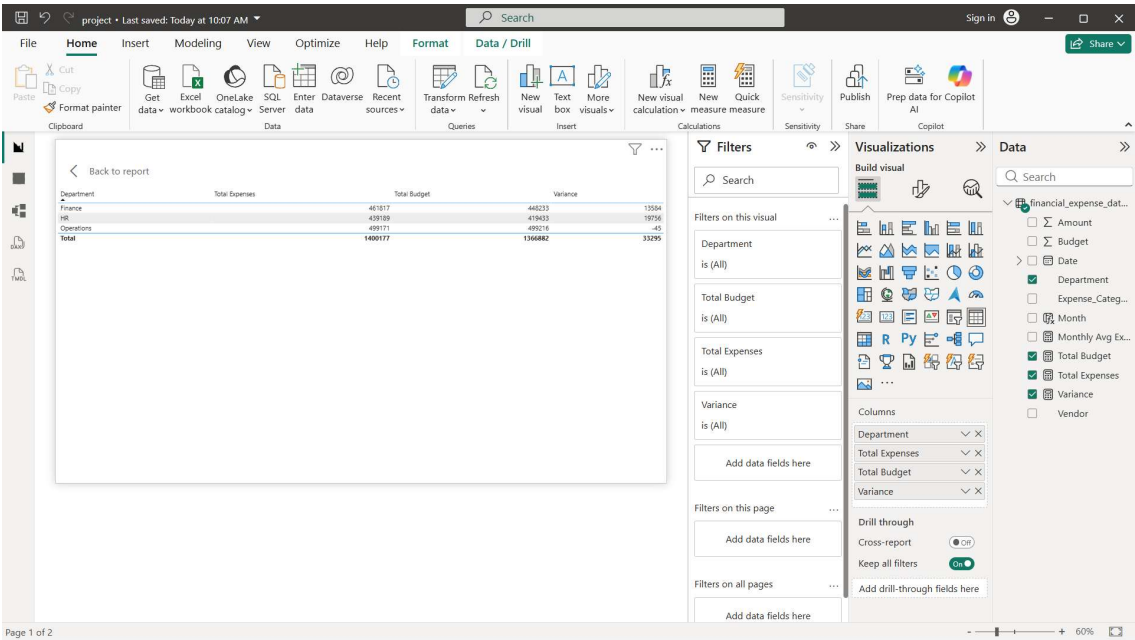
6. Compare actual expenses against budget using visuals

A clustered column chart was used to compare actual expenses with allocated budgets for each department. This visual clearly shows differences between planned and actual spending. The comparison helps identify budget utilization efficiency and reveals whether departments are managing finances within approved limits.



7. Identify departments exceeding budget limits

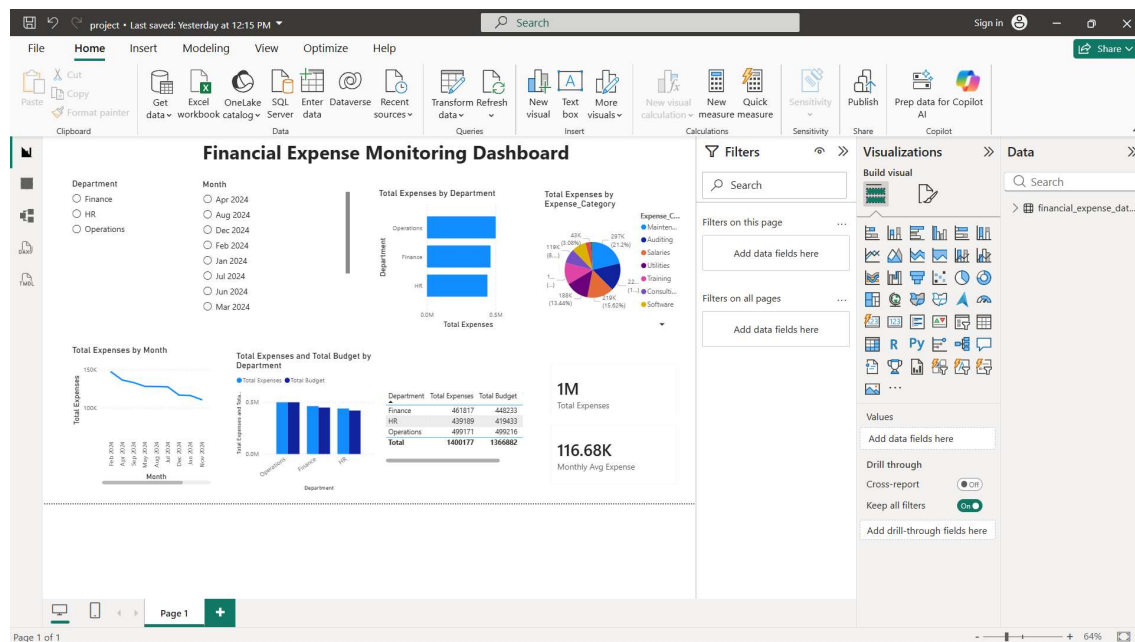
Using a table visualization and variance calculation (difference between actual expenses and budget), departments exceeding budget limits were identified. A positive variance indicates overspending. Highlighting these departments allows management to investigate causes and implement corrective actions.



8. Design a financial monitoring dashboard

A comprehensive dashboard was designed containing:

- Department expense bar chart
- Category distribution pie chart
- Monthly trend line chart
- Budget vs Actual comparison chart
- Variance table
- KPI cards for summary metrics
- Interactive slicers for filtering
- This enables quick decision-making and financial insight.



9. What cost-control actions can be recommended?

Based on analysis:

- Monitor high-spending departments regularly
- Optimize vendor contracts and service costs
- Set stricter budget tracking alerts
- Reduce non-essential operational expenses
- Improve forecasting using historical trends
- Implement periodic financial audits
- These actions help improve financial efficiency.

These actions can reduce unnecessary spending and improve financial sustainability.